SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage : 20%

QUESTIONS: **Total Marks: 50**

Annexure A

Scenario-Based

Code of Conduct:

I L. Infant Ency (1965260003) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Ency

Annexure A
Scenario-base
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1. Secure Data Transmission using Block Cipher
A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

- **A.** Explain why this issue occurs.
- **B.** Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System
A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- **A.** Identify the most suitable algorithm for this scenario.
- **B.** Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network
Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- **A.** Explain how the key exchange works in this scenario.
- **B.** Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security
An organization plans to implement RSA for secure email communication.

- **A.** Describe how RSA can be used for encryption and decryption in this scenario.
- **B.** Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices
A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

- **A.** Analyze the constraints of IoT devices.
- **B.** Recommend a suitable cryptographic approach and justify your answer.

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

CO2 AT2

Assessment Tool 2

Course name: Cryptography

Course Code: CSA5117

Name: L. Infant Ency

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1) Secure Data Transmission using Block Cipher:

a) Why this issue occurs:

=> The company uses ECB (Electronic codebook) mode to encrypt financial data

=> In ECB mode, identical plaintext blocks produce identical ciphertext blocks when the same key is used

=> This reveals patterns in the encrypted data, making it easier for attackers to identify information.

=> Therefore, ECB mode is not suitable for securely transmitting sensitive financial data

b) More secure block cipher mode: ~~and justify~~

=> CBC (Cipher Block Chaining) or GCM (Galois/counter mode) can be used instead of ECB

=> CBC uses an initialization vector (IV) to make identical Plaintext blocks produce different Ciphertext blocks

=> GCM provides both confidentiality and integrity by encryption data and detecting unauthorized modification

=> GCM is recommended because it provides strong security and helps detect tampering during transmission.

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4) Public key cryptography for email security:

a) How RSA can be used for encryption and decryption

=> RSA is an asymmetric encryption algorithm that uses a public key and a private key

=> The sender encrypts the email ~~is sent to the recipient through the network~~ using the recipient's public key

=> The encrypted email is sent to the recipient through the network

=> The recipient uses their private key to decrypt the email and read the original message

=> The ensures that only the intended recipient can decrypt the message, provided the private key remains secure

How key distribution and management play a role in maintaing Security:

⇒ Public key distribution: Public keys must be shared securely and verified using digital certificates

⇒ Private key Protection: The Private key must be kept secret and protected from unauthorized access

⇒ Key storage: key should be stored securely and accessed only by authorized users

⇒ Key renewal: Keys should be updated periodically to maintain Security

⇒ Key revocation: Compromised or invalid keys must be properly distributed, verified, stored and managed